Writing Equations from Word Situations
Answer Key

Grades 6–7   Expressions and Equations

Quick Answers

1. 12	4. 6	7. 84	10. 33
2. 175	5. 14	8. 48	
3. 17	6. 8	9. 6	

Curriculum

Level: Grades 6–7   Expressions and Equations
6.EE.A – *problems 3, 7*
6.EE.B.7 – *problems 1, 2, 6*
7.EE.B.4 – *problems 4, 5, 8, 9, 10*
Difficulty 1–4 of 5 across 10 tagged checks.

1. Pencils: 6 equal packs, 72 pencils in all.

Step 1. Name the unknown: let p be the number of pencils in one pack. Equal packs means repeated addition, so the situation multiplies.
Step 2. The equation is $6p = 72$. Undo the multiplication by dividing both sides by 6.
Step 3. $p = 72 \div 6$, so each pack holds $\boxed{\text{12 pencils}}$
Check: $6 \times 12 = 72$ pencils, which is what the story says.

2. Club fund: spent 45, left with 130.

Step 1. The unknown is the amount before the purchase, so let x be that amount in dollars. Spending subtracts.
Step 2. The equation is $x - 45 = 130$. Undo the subtraction by adding 45 to both sides.
Step 3. $x = 130 + 45$, so the fund held $\boxed{\text{175 dollars}}$
Listen for: 85, from subtracting instead of adding. Ask which number is the larger amount — the fund before spending has to be more than what is left.

3. Taxi: flat 3 plus 2 per mile, ride of 7 miles.

Step 1. One amount happens once (the flat charge) and one happens per mile, so the rate multiplies the number of miles and the flat charge is added on.
Step 2. The expression is $3 + 2m$ dollars for m miles. Substitute $m = 7$.
Step 3. $3 + 2(7) = 3 + 14$, so the fare is $\boxed{\text{17 dollars}}$
Common wrong answer: 35 — the flat charge was included in the per-mile rate, giving $(3 + 2)(7)$. The flat charge is paid once, not once per mile.

4. Gym: 20 joining fee plus 15 a month, 110 paid.

Step 1. Let n be the number of months. The joining fee is paid once and the monthly charge repeats, so the total is a one-time amount plus a rate times n.
Step 2. The equation is $20 + 15n = 110$. Undo the addition first: $15n = 90$.
Step 3. Divide by the rate: $n = 90 \div 15$, so Ren has been a member for $\boxed{\text{6 months}}$
Check: $20 + 15(6) = 110$ dollars.

5. Garden bed: perimeter 46 cm, width 9 cm.

Step 1. Let x be the length in cm. A rectangle has two lengths and two widths, so the perimeter is $2x + 2(9)$.
Step 2. The equation is $2x + 18 = 46$. Subtract the two widths: $2x = 28$.
Step 3. Divide by 2: the length is $\boxed{\text{14 cm}}$
Listen for: $x = 23$, from halving the perimeter and forgetting the width, or $x = 37$, from subtracting one width instead of two.

6. Oats: $\frac{3}{4}$ cup per batch, 6 cups used.

Step 1. Let n be the number of batches. Each batch uses the same amount, so the situation is a rate times a count.
Step 2. The equation is $\frac{3}{4}n = 6$. Undo multiplication by a fraction by multiplying by its reciprocal, $\frac{4}{3}$.
Step 3. $n = 6 \cdot \frac{4}{3} = \frac{24}{3}$, so the baker made $\boxed{\text{8 batches}}$
Teaching note: the answer is larger than 6, which surprises students. Each batch uses less than a cup, so more than six batches fit into six cups.

7. Pool: 300 gallons, draining 12 gallons per minute.

Step 1. The pool starts full and loses a fixed amount every minute, so the rate multiplies the time and is subtracted from the starting amount.
Step 2. The expression is $300 - 12t$ gallons after t minutes. Substitute $t = 18$.
Step 3. $12 \times 18 = 216$, and $300 - 216$ leaves $\boxed{\text{84 gallons}}$
Common wrong answer: 288 — the drain rate was subtracted once instead of once per minute.

8. Bill: 6 coupon, then split three ways, each paying 14.

Step 1. Let x be the bill in dollars before the coupon. The story happens in order: the coupon comes off first, then the rest is divided by three.
Step 2. The equation is $\dfrac{x-6}{3} = 14$. Undo the division first by multiplying both sides by 3: $x - 6 = 42$.
Step 3. Add the coupon back: the bill was $\boxed{\text{48 dollars}}$
Listen for: $x = 36$, from taking the coupon off after splitting. Order matters, and the equation records the order.

9. Error analysis: skates cost 5 plus 3 per hour, total 23.

Step 1. Ask what each number does. The 5 is paid once, no matter how long Ana skates; the 3 is paid for each hour. Ben's equation $5h + 3 = 23$ multiplies the one-time charge by the hours and adds the hourly rate once — it has the two roles swapped.
Step 2. The correct model attaches the variable to the rate: $5 + 3h = 23$. Subtract the flat charge: $3h = 18$.
Step 3. Divide by the rate: Ana skated for $\boxed{\text{6 hours}}$
What to say: read the equation back as a sentence. Ben's says "five dollars for each hour, plus three dollars once", which is not what the sign at the rink says.

10. Coding club: two thirds signed up, 4 dropped out, 18 left.

Step 1. Let s be the number of members. Two thirds of them is $\frac{2}{3}s$, and the story then removes 4 from that group.
Step 2. The equation is $\frac{2}{3}s - 4 = 18$. Undo the subtraction first: $\frac{2}{3}s = 22$.
Step 3. Multiply by the reciprocal $\frac{3}{2}$: the club has $\boxed{\text{33 students}}$
Check: two thirds of 33 is 22 contestants, and $22 - 4 = 18$ after the dropouts.